DE–6221 **11**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2013.

MATHEMATICAL FOUNDATION OF COMPUTER SCIENCE

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — ($10 \times 3 = 30$ marks)

Answer ALL the questions.

1. What is Bi-conditional statement? Give the truth table of the Bi-conditional statement.
2. Define PDNF.
3. What is universal set? Give an example.
4. Define partitions.
5. What is bijective? Give an example.
6. What are binary and n-ary operations?
7. What is Algebraic system?
8. Define Groups.
9. What is multigraph? Give an example.
10. What is reachability matrix?

SECTION B — ($4 \times 10 = 40$ marks)

Answer any FOUR questions.

11. Explain any three connectives with examples.
12. Construct the truth table for $\neg(PVQ) \cap (PVR)$.

13. Let R be the relation represented by the matrix.

$$M_R = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$ Find the matrix representing

(a) R^{-1}

(b) R^2

14. Let $f : R \to R$ and $g : R \to R$ where R is the set of real numbers find $f_o g$ and $g_o f$ if $f(x) = x^2 - 2$ and $g(x) = x + 4$.

15. Prove the set of all semi group endomorphism of the semi group is a semi group under the operation of left composition.

16. Prove sum of the degree of the points of a graph G is twice the number of lines i.e. $\sum \deg v_i = 2q$. (v = vertices, q = edges).

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Obtain the principal disjunctive normal form of $(P(P \cap Q)V(\neg P \cap R)V(Q \cap R)$ Also find PCNF.

18. Show that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ using characteristic function.

19. State and prove the Lagrange's theorem.

DE–6222 **12**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2013.

COMPUTER ARCHITECTURE

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. What are different kinds of addressing modes?

2. Illustrate any one set of operands with operator used for media and signal processing.

3. Mention any two limitations of ILP.

4. Distinguish dynamic scheduling from static scheduling.

5. What is meant by coarse grain parallelism?

6. List down static multiple issues of parallelism.

7. Identify the types of storage devices.

8. Mention the various parameters of I/O performance.

9. Mention the different types of synchronisation models.

10. What is a thread level parallelism?

SECTION B — ($4 \times 10 = 40$ marks)

Answer any FOUR questions.

11. Give a brief account on principles of computer design.

12. Discuss the concept of linear pipeline processors.

13. Write about VLIW architecture with neat diagram.

14. Elucidate the concept of cache memory and also write its importance.

15. Narrate the procedure of system control and I/O processors of a system organization.

16. Give a note on multithreading mechanism.

SECTION C — ($2 \times 15 = 30$ marks)

Answer any TWO questions.

17. Explain with neat diagram, the various classification of instruction set architecture.

18. With neat diagram explain the mechanism followed in IA 64 and Itanium Processor.

19. Give a detailed description on distributed shared memory architectures with neat diagram.

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DE–6223 **13**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2013.

DATA STRUCTURES USING C++

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL the questions.

1. Write notes on constructor.

2. What is a class? Give an example.

3. Write down the applications of Queue.

4. What is a Stack? What are the operations on stack?

5. Write down the steps to evaluate a given Postfix Expression.

6. What is meant by doubly linked list?

7. List out the Uses of Tree.

8. Write the steps to traverse a binary Tree in pre order.

9. Write notes on Sequential searching.

10. What is an insertion sort?

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. How to define and declare a two dimensional array? Explain with an example.

12. Explain infix and prefix concept with examples.

13. Explain the procedure to delete a node at the end of the linked list.

14. Discuss about quick sort with an example.

15. Explain the concept of Binary Search.

16. Sort the following array using insertion sort: 25, 15, 30, 9, 99, 20, 26.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Explain the implementation of queue operations with examples.

18. Discuss the various binary Tree representation.

19. Describe the Quick sort with an example.

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DE–6224 **14**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2013.

VISUAL PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Define variable. Write any four Valid and Invalid Variables.

2. Define Database.

3. Define Module.

4. Mention the role of database manager.

5. What is an object?

6. Write note on OLE.

7. Define Exception Handling.

8. What is MDI?

9. Define files.

10. What is DLL?

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain Data types in VB with suitable example.

12. Explain procedures and functions.

13. Write in detail about Menus in VB.

14. What are the components of VC++?

15. Write a program in VB to Reverse a given set of N numbers.

16. Explain the concept of object linking and embedding in detail with example.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Write a VC++ program for database access through ODBC.

18. Explain the features of software development kit tools in detail.

19. Describe the toolbox control.

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DE–6225 **15**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2013.

DATA BASE MANAGEMENT SYSTEM

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Define: Data base management system.

2. Distinguish internal level from external level.

3. Define:

   (a) Super key

   (b) Primary key

   (c) Foreign key

4. With an illustration, state the project operation.

5. Define row triggers and statement triggers.

6. Identify different types of segments used in oracle table space.

7. List down SQL built in domain types.

8. Write the statement for union operation. Illustrate.

9. Define the term, "Dynamic SQL".

10. What are database objects?

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Illustrate and explain the schema and subschema of a database.

12. Explain the division and assignment operations used in relational algebra with examples.

13. Write down the overview and benefits of oracle.

14. Briefly explain the basic structure of SQL queries.

15. Discuss the importance of complex queries used in SQL

16. Narrate the procedure of embedded SQL.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Give a detailed description on data manipulation language and data definition language.

18. Describe the oracle database storage and indexing procedure.

19. Explain the query processing and optimization techniques followed in oracle.

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DE–6226 **21**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2013.

COMPUTER NETWORKS

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Write short note on WAN.

2. Compare OSI and TCP reference model.

3. What is framing?

4. What are error correcting codes?

5. How tunneling works?

6. Mention various services provided to transport layer.

7. Give various elements of transport protocol.

8. What is meant by crash recovery?

9. Discuss about WWW.

10. Give any three services of electronic mail.

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Discuss any two transmission media.

12. What is ALOHA? Explain.

13. Write a detail note on hierarchical routing.

14. Give a brief note on TCP Congestion control.

15. What is DNS? Explain.

16. Discuss about Secret Key algorithm.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Explain about a one bit sliding window protocol.

18. Discuss any three routing algorithms.

19. What is data compression? Explain.

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DE–6227 **22**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2013.

SOFTWARE ENGINEERING

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Define : Software Engineering.

2. Write a note on : The Spiral Model.

3. What is software scope?

4. What are Project and Technical Risks?

5. Define : Timeline charts.

6. Write a note on : Cost of Quality.

7. List down Analysis Principles.

8. What is Stepwise Refinement?

9. Define : Cyclomatic Complexity.

10. What is Regression Testing?

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Explain briefly on : Project Management concepts.

12. Write short notes on : Software Project Estimation.

13. Explain software risk identification.

14. Discuss the Basic Concepts in project Scheduling and Tracking.

15. Write short notes on : Design Principles.

16. Explain about Validation Testing.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Explain about Decomposition Techniques.

18. Describe in detail, software configuration management.

19. Discuss the following :

    (a) Transform Mapping (8)

    (b) Unit Testing. (7)

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DE–6228 **23**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2013.

INTERNET PROGRAMMING AND WEB DESIGN

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL the questions.

1. Write an overview of an Internet Programming.

2. Write down the steps involved for developing Intranet Applications.

3. Define Java.

4. What is meant by PERL?

5. Write notes on VBScript.

6. What are the other Scripting Languages?

7. Define HTML.

8. Write a note on Shokwave and Lingo.

9. Write a note on VDOlive Technology.

10. What is meant by Dynamic Graphics?

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Discuss about the Security and Encryption of Internet Programming.

12. Discuss about the Java Script.

13. Write in detail about the HTML Tags.

14. Write in detail about creating a custom Integrated Application with Multiple Protocols.

15. Discuss in detail about animation in web graphics.

16. Write a note on Pulling Web Information.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Describe the java in windows in detail.

18. Explain briefly about Creating Netscape Navigator Plug-Ins.

19. Explain in detail about the Images and Hyperlinks in web graphics.

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DE–6229 **24**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2013.

IMAGE PROCESSING AND ANALYSIS

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL questions.

1. Define the term scenes and images.

2. What is acuity and contour?

3. What are the various spatial domain techniques?

4. What is image averaging?

5. What are the two broad groups of error criterion?

6. What is Karhunen–Loeve (K–L) transform?

7. How can region be extracted?

8. What are the steps involved in the process of generation of edge map?

9. Define the term interior Boundary.

10. What is Genus of an image?

SECTION B — ($4 \times 10 = 40$ marks)

Answer any FOUR questions.

11. Write in detail about the brightness adaptation and contrast with diagram.

12. Discuss about the high–pass filtering and homomorphic filtering.

13. Describe restoration by maximum a posterior estimation.

14. Discuss the plane–to–plane transformation in detail.

15. Write in detail about the edge elements extraction by thresholding.

16. Explain clustering with algorithms.

SECTION C — ($2 \times 15 = 30$ marks)

Answer any TWO questions.

17. Explain image sharpening in detail with example.

18. Discuss about the Lossy compression.

19. Discuss about the pattern fitting approach.

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DE–6230 **25**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2013.

OPERATING SYSTEMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

SECTION A — (10 × 3 = 30 marks)

Answer ALL the questions.

1. What are Distributed Systems?

2. Write a note on: System Programs.

3. What are process states?

4. Define: Monitors.

5. What is Contiguous Memory Allocation?

6. Write a short note on: Thrashing.

7. Write about Directory Structure.

8. What is Recovery?

9. What are Distributed File Systems?

10. Define: Concurrency Control.

SECTION B — (4 × 10 = 40 marks)

Answer any FOUR questions.

11. Discuss about Clustered Systems and Hand held systems.

12. Write short notes on: System calls.

13. Discuss about Deadlock Avoidance.

14. Explain Page Replacement in Virtual Memory.

15. Discuss briefly about Disk Scheduling.

16. Write short notes on: Linux OS.

SECTION C — (2 × 15 = 30 marks)

Answer any TWO questions.

17. Explain about Operating System Services.

18. Discuss in detail, Multithreading Models.

19. Write about the following:

    (a) Segmentation in Memory Management. (8)

    (b) Remote File Access. (7)

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DE–6643 **16**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION, MAY 2013.

C++ AND DATA STRUCTURES — LAB

(2007 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select and give ONE question to each candidate by LOT system.

Break-up of marks

Record	: 10
Algorithm	: 10
Program	: 50
Debugging	: 10
Execution	: 10
Result	: 10
Total	: 100

1. (a) Write a C++ program to convert a given number into words for number 1 to 5.

   (b) Write a program using polymorphism to calculate the square of any two numbers of type int, float, double and long.

---------- Cut here ----------

2. (a) Write a C++ program to find the roots of a quadratic equation $ax^2 + bx + c = 0$.

   (b) Write a C++ program to implement push and pop operation on stack.

3. (a) Write a C++ program to find the roots of a quadratic equation using function.

   (b) Write a C++ program to evaluate the given mathematical expression using stack.

---------- Cut here ----------

4. (a) Write a C++ program for matrix multiplication.

   (b) Write a C++ program to implement insert and delete operations on linked list structure.

---------- Cut here ----------

5. (a) Write a C++ program to find the factorial of a given number using recursion.

   (b) Write a C++ program to implement insert and delete operations on queue using array concept.

---------- Cut here ----------

6. (a) Write a C++ program to read TEN values to an array variable. Use pointers to locate and display each value.

   (b) Write a C++ program to implement Insert and delete operations on Queue using array concept.

---------- Cut here ----------

7. (a) Using dynamic constructors write a C++ program to concatenate two given strings.

   (b) Write a C++ program for linked list implementation of queue operations.

8. (a) Write a menu driven program in C++ to add and subtract given two matrices of order m × n defined in class, using operator overloading.

(b) Write a program in C++ to sort a set of elements using selection sort.

---------- Cut here ----------

9. (a) Write a C++ program to implement push and pop operations on stack.

(b) Write a C++ program to sort a set of elements using Quick Sort.

DE–6644 **17**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION,
MAY 2013.

VISUAL PROGRAMMING – LAB

(2007 Onwards)

Time : Three hours

Maximum : 100 marks
Record : 10
Algorithm : 10
Program : 50
Debugging : 10
Execution : 10
Result : 10
Total : 100

Students has to answer one question by lot system.

1. (a) Write an event procedure for the following
   (i) Reverse a string.
   (ii) Determine whether the given string is palindrome or not.
   (iii) Change the case of a string to upper of lower (use Label, Textbox, CommandButton).

   (b) Write a visual C++ program to fill background of client area with bitmap.

---------- Cut here ----------

2. (a) Using MSflex grid control, display the multiplication and addition table of 20 rows and colums (Use MSflex Grid, Button)

   (b) Write a visual C++ program to find out whether a mouse is attached or not and if attached how many buttons are present or not.

3. (a) Using a control array, create a simple calculator which will do the following operation

   (i) Addition

   (ii) Subtraction

   (iii) Multiplication

   (iv) Division

   (v) Square

   (vi) Square root

   (vii) Modulus

   (viii) Power

   (ix) Percentage.

   (Using Textbox, CommondButtons, Frames)

   (b) Write a Visual C++ program to create a list box in a window.

---------- Cut here ----------

4. (a) Create a VB application with a DriveList Box, Dirlistbox.

   (i) Select exe file and execute it

   (ii) Select the picture and load it to the form

   (iii) Filter the file in the file list box according to the extension chosen in the combo box.

   (b) Write a visual C++ program to handle windows messages in MFC program.

5. (a) Using Ax control create a textbox that accepts only numeric value with the following properties

   (i) Background of the textbox

   (ii) Foreground of the text box

   (iii) Text property of the text box

   (iv) Resize the textbox at the standard application without using properties.

   (b) Write a Visual C++ program to create a list box in a window.

---------- Cut here ----------

6. (a) Using build in Ax controls, develop the windows NOTEPAD with file and edit

   menu operation and also display the floating menu whenever necessary.

   (Use Richtext box, Menu editor).

   (b) Write a Visual C++ program to create a window of desired size using MFC.

---------- Cut here ----------

7. (a) Write a VB program to perform the following operation in a record of random Access file.

   (i) Insert

   (ii) Delete

   (iii) Search.

   (b) Write a Visual C++ program to create a List box in a window.

8. (a) Using ActiveX control create a textbox that accepts only numeric value with the following properties

   (i) Background of the textbox

   (ii) Foreground of the textbox

   (iii) Text property of the text box

   (iv) Resize the textbox at the standard application without using properties.

   (b) Write a Visual C++ program to handle windows messages in MFC program.

---------- Cut here ----------

9. (a) Create a table hospital with the following fields patient number, patient name, address, blood group, disease.

   Using Ax data access object, develop a hospital mgnt. system with following operation

   (i) Insert

   (ii) Search

   (iii) Modify

   (iv) Display.

   (b) Write a visual C++ program to create a window of desired size using MFC.

---------- Cut here ----------

10. (a) Using MSflex grid control, display the multiplication and addition table of 20 rows and columns (Use MSflex grid, Button).

    (b) Write a Visual C++ program to handle windows messages in MFC program.

DE–6645 **26**

DISTANCE EDUCATION

M.Sc. (Computer Science) DEGREE EXAMINATION, MAY 2013.

INTERNET PROGRAMMING — LAB III

(2007 onwards)

Time : Three hours Maximum : 100 marks

Students has to answer one question by lot system.

Break up marks :

Record	: 10
Algorithm	: 10
Program	: 50
Debugging	: 10
Execution	: 10
Result	: 10
Total	: 100

1. (a) Write a Java Script to create a student request form for joining any one of the Extra Curricular activities (Sports, NSS, NCC, Fine Arts).

   (b) Design and publish a web page for a college.

---------- cut here ----------

2. (a) Write a Java program using Applet to display the different colors and fonts.

   (b) Write a swing program to create the Tabbed panels.

---------- cut here ----------

3. (a) Write a VB script to display the calender for a given month and year.

   (b) Design a web page using HTML to give information about various homes and their rates in the Apartment buildings in Chennai.

4. (a) Develop a single page advertisement for a shop to be opened shortly.

   (b) Write a menu drive Java program to do database functions using JDBC connectivity facility.

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5. (a) Write a Java Script to create a window by using the confirm message.

   (b) Write a swing program to create the Tabbed Panels.

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6. (a) Write a java script to create Order form to select the house articles.

   (b) With a VB script to do the following :

      (i) Check the given password.

      (ii) Change the existing password.

———————————————— cut here ————————————————

7. (a) Develop and publish a webpage for a college.

   (b) Write a Java program to create a color as the Back Ground.

———————————————— cut here ————————————————

8. (a) Create a web page in the format of front page of a news paper using Text links. Align the text with colors.

   (b) Write a swing program to create the Tabbed panels.

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9. (a) Write a VB script to create a calendar for given month and year.

   (b) Write a Java script to create a order form to select the house articles.

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